Adoption of Semantic Operators. Based on a preprint [54] of this work, Google recently implemented experimental semantic operators in BigQuery Dataframes [12], including `sem_map`, `sem_filter`, `sem_join`, `sem_agg`, `sem_topk`, `sem_search`, and `sem_sim_join`.

7 CONCLUSION

In this work, we proposed semantic operators for bulk semantic processing and provide the first formalized model for general-purpose, language-based transformations over data. Our results across a diverse set of applications, including fact-checking, biomedical multi-label classification, search, and topic analysis, demonstrate the generality, expressiveness and robustness of the semantic operator model as well as our optimization approach. For each task, we find that LOTUS programs capture state-of-the-art AI applications with low development overhead, match or exceed result quality of recent LLM-powered analytics systems, and substantially reduce cost, while ensuring accuracy guarantees. Since open-sourcing LOTUS, we have seen a growing user-base, as well as adoption of semantic operators among large database vendors. We believe both represent exciting steps towards new data systems that integrate powerful AI reasoning capabilities over vast knowledge corpora.

ACKNOWLEDGMENTS

This research was supported in part by affiliate members and other supporters of the Stanford DAWN project, including Meta, Google, and VMware, as well as Cisco, SAP, and a Sloan Fellowship. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the sponsors.

REFERENCES

[1] [n.d.]. AI Functions on Databricks. https://docs.databricks.com
[2] [n.d.]. arXiv.org ePrint archive. https://arxiv.org
[3] [n.d.]. Custom Search JSON API | Programmable Search Engine. https://developers.google.com/custom-search/v1/overview
[4] [n.d.]. Discovery Insight Platform. https://www.findourview.com
[5] [n.d.]. GAIR-NLP/factool: FacTool: Factuality Detection in Generative AI. https://github.com/GAIR-NLP/factool
[6] [n.d.]. Introducing Meta Llama 3: The most capable openly available LLM to date. https://ai.meta.com/blog/meta-llama-3/
[7] [n.d.]. LangChain. https://www.langchain.com/
[8] [n.d.]. Large Language Model (LLM) Functions (Snowflake Cortex) | Snowflake Documentation. https://docs.snowflake.com/user-guide/snowflake-cortex/llm-functions
[9] [n.d.]. LLM with Vertex AI only using SQL queries in BigQuery. https://cloud.google.com/blog/products/ai-machine-learning/llm-with-vertex-ai-only-using-sql-queries-in-bigquery
[10] [n.d.]. OpenAI Platform. https://platform.openai.com
[11] [n.d.]. pandas - Python Data Analysis Library. https://pandas.pydata.org/
[12] [n.d.]. python-bigquery-dataframes/notebooks/experimental/semantic_operators.ipynb at main · googleapis/python-bigquery-dataframes. https://github.com/googleapis/python-bigquery-dataframes/blob/main/notebooks/experimental/semantic_operators.ipynb
[13] [n.d.]. Querying - LlamaIndex 0.9.11.post1. https://docs.llamaindex.ai/en/stable/understanding/querying/querying.html
[14] 2023. Large Language Models for sentiment analysis with Amazon Redshift ML (Preview) | AWS Big Data Blog. https://aws.amazon.com/blogs/big-data/large-language-models-for-sentiment-analysis-with-amazon-redshift-ml-preview/ Section: Amazon Redshift.
[15] 2024. mixedbread-ai/mxbai-rerank-large-v1 · Hugging Face. https://huggingface.co/mixedbread-ai/mxbai-rerank-large-v1
[16] Griffin Adams, Alexander Fabbri, Faisal Ladhak, Eric Lehman, and Noémie Elhadad. 2023. From Sparse to Dense: GPT-4 Summarization with Chain of Density Prompting. http://arxiv.org/abs/2309.04269 arXiv:2309.04269 [cs].
[17] Eric Anderson, Jonathan Fritz, Austin Lee, Bohou Li, Mark Lindblad, Henry Lindeman, Alex Meyer, Parth Parmar, Tanvi Ranade, Mehul A. Shah, Benjamin Sowell, Dan Tecuci, Vinayak Thapliyal, and Matt Welsh. 2024. The Design of an LLM-powered Unstructured Analytics System. https://doi.org/10.48550/arXiv.2409.00847 arXiv:2409.00847 [cs].
[18] Simran Arora, Brandon Yang, Sabri Eyuboglu, Avanika Narayan, Andrew Hojel, Immanuel Trummer, and Christopher Ré. 2023. Language Models Enable Simple Systems for Generating Structured Views of Heterogeneous Data Lakes. http://arxiv.org/abs/2304.09433 arXiv:2304.09433 [cs].
[19] Asim Biswal, Liana Patel, Siddarth Jha, Amog Kamsetty, Shu Liu, Joseph E. Gonzalez, Carlos Guestrin, and Matei Zaharia. 2024. Text2SQL is Not Enough: Unifying AI and Databases with TAG. https://doi.org/10.48550/arXiv.2408.14717 arXiv:2408.14717 [cs].
[20] Mark Braverman and Elchanan Mossel. [n.d.]. Noisy sorting without resampling. ([n. d.]).
[21] Yapei Chang, Kyle Lo, Tanya Goyal, and Mohit Iyyer. 2024. BooookScore: A systematic exploration of book-length summarization in the era of LLMs. http://arxiv.org/abs/2310.00785 arXiv:2310.00785 [cs].
[22] Lingjiao Chen, Matei Zaharia, and James Zou. 2023. FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance. http://arxiv.org/abs/2305.05176 arXiv:2305.05176 [cs].
[23] I.-Chun Chern, Steffi Chern, Shiqi Chen, Weizhe Yuan, Kehua Feng, Chunting Zhou, Junxian He, Graham Neubig, and Pengfei Liu. 2023. FacTool: Factuality Detection in Generative AI – A Tool Augmented Framework for Multi-Task and Multi-Domain Scenarios. https://doi.org/10.48550/arXiv.2307.13528 arXiv:2307.13528 [cs].
[24] E F Codd. 1970. A Relational Model of Data for Large Shared Data Banks. 13, 6 (1970).
[25] Hanjun Dai, Bethany Yixin Wang, Xingchen Wan, Bo Dai, Sherry Yang, Azade Nova, Pengcheng Yin, Phitchaya Mangpo Phothilimthana, Charles Sutton, and Dale Schuurmans. 2024. UQE: A Query Engine for Unstructured Databases. https://doi.org/10.48550/arXiv.2407.09522 arXiv:2407.09522 [cs].
[26] Sanjoy Dasgupta. [n.d.]. The hardness of k-means clustering. ([n. d.]).
[27] Shrey Desai and Greg Durrett. 2020. Calibration of Pre-trained Transformers. https://arxiv.org/abs/2003.07892v3
[28] Karel D'Oosterlinck, Omar Khattab, François Remy, Thomas Demeester, Chris Develder, and Christopher Potts. 2024. In-Context Learning for Extreme Multi-Label Classification. https://doi.org/10.48550/arXiv.2401.12178 arXiv:2401.12178 [cs].
[29] Karel D'Oosterlinck, François Remy, Johannes Deleu, Thomas Demeester, Chris Develder, Klim Zaporojets, Aneiss Ghodsi, Simon Ellershaw, Jack Collins, and Christopher Potts. 2023. BioDEX: Large-Scale Biomedical Adverse Drug Event Extraction for Real-World Pharmacovigilance. https://doi.org/10.48550/arXiv.2305.13395 arXiv:2305.13395 [cs].
[30] Matthijs Douze, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Gergely Szilvasy, Pierre-Emmanuel Mazaré, Maria Lomeli, Lucas Hosseini, and Hervé Jégou. 2024. The Faiss library. https://doi.org/10.48550/arXiv.2401.08281 arXiv:2401.08281 [cs].
[31] Andrew Drozdov, Honglei Zhuang, Zhuyun Dai, Zhen Qin, Razieh Rahimi, Xuanhui Wang, Dana Alon, Mohit Iyyer, Andrew McCallum, Donald Metzler, and Kai Hui. 2023. PaRaDe: Passage Ranking using Demonstrations with Large Language Models. https://doi.org/10.48550/arXiv.2310.14408 arXiv:2310.14408 [cs].
[32] Joe Hellerstein, Christopher Ré, Florian Schoppmann, Daisy Zhe Wang, Eugene Fratkin, Aleksander Gorajek, Kee Siong Ng, Caleb Welton, Xixuan Feng, Kun Li, and Arun Kumar. 2012. The MADlib Analytics Library or MAD Skills, the SQL. http://arxiv.org/abs/1208.4165 arXiv:1208.4165 [cs].
[33] C. A. R. Hoare. 1961. Algorithm 65: find. *Commun. ACM* 4, 7 (July 1961), 321–322. https://doi.org/10.1145/366622.366647
[34] Jeff Johnson, Matthijs Douze, and Hervé Jégou. 2017. Billion-scale similarity search with GPUs. http://arxiv.org/abs/1702.08734 arXiv:1702.08734 [cs].
[35] Daniel Kang, Peter Bailis, and Matei Zaharia. 2019. BlazeIt: Optimizing Declarative Aggregation and Limit Queries for Neural Network-Based Video Analytics. http://arxiv.org/abs/1805.01046 arXiv:1805.01046 [cs].
[36] Daniel Kang, John Emmons, Firas Abuzaid, Peter Bailis, and Matei Zaharia. 2017. NoScope: Optimizing Neural Network Queries over Video at Scale. http://arxiv.org/abs/1703.02529 arXiv:1703.02529 [cs].
[37] Daniel Kang, Edward Gan, Peter Bailis, Tatsunori Hashimoto, and Matei Zaharia. 2020. Approximate selection with guarantees using proxies. *Proceedings of the VLDB Endowment* 13, 12 (Aug. 2020), 1990–2003. https://doi.org/10.14778/3407790.3407804
[38] Daniel Kang, Edward Gan, Peter Bailis, Tatsunori Hashimoto, and Matei Zaharia. 2022. Approximate Selection with Guarantees using Proxies. https://doi.org/10.48550/arXiv.2004.00827 arXiv:2004.00827 [cs].
[39] Daniel Kang, John Guibas, Peter Bailis, Tatsunori Hashimoto, Yi Sun, and Matei Zaharia. 2021. Accelerating approximate aggregation queries with expensive predicates. *Proceedings of the VLDB Endowment* 14, 11 (July 2021), 2341–2354. https://doi.org/10.14778/3476249.3476285
[40] Daniel Kang, John Guibas, Peter Bailis, Tatsunori Hashimoto, and Matei Zaharia. [n.d.]. Task-agnostic Indexes for Deep Learning-based Queries over Unstructured Data. ([n. d.]).

[41] Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. 2023. DSPy: Compiling Declarative Language Model Calls into Self-Improving Pipelines. https://arxiv.org/abs/2310.03714v1

[42] Omar Khattab and Matei Zaharia. 2020. ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT. https://arxiv.org/abs/2004.12832v2

[43] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient Memory Management for Large Language Model Serving with PagedAttention. https://arxiv.org/abs/2309.06180v1

[44] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2021. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. https://doi.org/10.48550/arXiv.2005.11401 arXiv:2005.11401 [cs].

[45] Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Christian Cosgrove, Christopher D. Manning, Christopher Ré, Diana Acosta-Navas, Drew A. Hudson, Eric Zelikman, Esin Durmus, Faisal Ladhak, Frieda Rong, Hongyu Ren, Huaxiu Yao, Jue Wang, Keshav Santhanam, Laurel Orr, Lucia Zheng, Mert Yuksekgonul, Mirac Suzgun, Nathan Kim, Neel Guha, Niladri Chatterji, Omar Khattab, Peter Henderson, Qian Huang, Ryan Chi, Sang Michael Xie, Shibani Santurkar, Surya Ganguli, Tatsunori Hashimoto, Thomas Icard, Tianyi Zhang, Vishrav Chaudhary, William Wang, Xuechen Li, Yifan Mai, Yuhui Zhang, and Yuta Koreeda. 2022. Holistic Evaluation of Language Models. https://arxiv.org/abs/2211.09110v2

[46] Yiming Lin, Madelon Hulsebos, Ruiying Ma, Shreya Shankar, Sepanta Zeigham, Aditya G. Parameswaran, and Eugene Wu. 2024. Towards Accurate and Efficient Document Analytics with Large Language Models. http://arxiv.org/abs/2405.04674 arXiv:2405.04674 [cs].

[47] Chunwei Liu, Matthew Russo, Michael Cafarella, Lei Cao, Peter Baille Chen, Zui Chen, Michael Franklin, Tim Kraska, Samuel Madden, and Gerardo Vitagliano. 2024. A Declarative System for Optimizing AI Workloads. http://arxiv.org/abs/2405.14696 arXiv:2405.14696 [cs].

[48] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2023. Lost in the Middle: How Language Models Use Long Contexts. https://doi.org/10.48550/arXiv.2307.03172 arXiv:2307.03172 [cs].

[49] Shu Liu, Asim Biswal, Audrey Cheng, Xiangxi Mo, Shiyi Cao, Joseph E. Gonzalez, Ion Stoica, and Matei Zaharia. 2024. Optimizing LLM Queries in Relational Workloads. http://arxiv.org/abs/2403.05821 arXiv:2403.05821 [cs].

[50] Shicheng Liu, Jialiang Xu, Wesley Tjangnaka, Sina J. Semnani, Chen Jie Yu, and Monica S. Lam. 2024. SUQL: Conversational Search over Structured and Unstructured Data with Large Language Models. https://doi.org/10.48550/arXiv.2311.09818 arXiv:2311.09818 [cs].

[51] Yao Lu, Aakanksha Chowdhery, Srikanth Kandula, and Surajit Chaudhuri. 2018. Accelerating Machine Learning Inference with Probabilistic Predicates. In *Proceedings of the 2018 International Conference on Management of Data (SIGMOD '18)*. Association for Computing Machinery, New York, NY, USA, 1493–1508. https://doi.org/10.1145/3183713.3183751

[52] Xueguang Ma, Xinyu Zhang, Ronak Pradeep, and Jimmy Lin. 2023. Zero-Shot Listwise Document Reranking with a Large Language Model. https://arxiv.org/abs/2305.02156v1

[53] MotherDuck. [n.d.]. Introducing the prompt() Function: Use the Power of LLMs with SQL! - MotherDuck Blog. https://motherduck.com/blog/sql-llm-prompt-function-gpt-models/

[54] Liana Patel, Siddharth Jha, Carlos Guestrin, and Matei Zaharia. 2024. LOTUS: Enabling Semantic Queries with LLMs Over Tables of Unstructured and Structured Data. https://doi.org/10.48550/arXiv.2407.11418 arXiv:2407.11418 [cs] version: 1.

[55] Liana Patel, Peter Kraft, Carlos Guestrin, and Matei Zaharia. 2024. ACORN: Performant and Predicate-Agnostic Search Over Vector Embeddings and Structured Data. *Proceedings of the ACM on Management of Data* 2, 3 (May 2024), 120:1–120:27. https://doi.org/10.1145/3654923

[56] Ronak Pradeep, Sahel Sharifymoghaddam, and Jimmy Lin. 2023. RankVicuna: Zero-Shot Listwise Document Reranking with Open-Source Large Language Models. http://arxiv.org/abs/2309.15088 arXiv:2309.15088 [cs].

[57] Ronak Pradeep, Sahel Sharifymoghaddam, and Jimmy Lin. 2023. RankZephyr: Effective and Robust Zero-Shot Listwise Reranking is a Breeze! http://arxiv.org/abs/2312.02724 arXiv:2312.02724 [cs].

[58] Zhen Qin, Rolf Jagerman, Kai Hui, Honglei Zhuang, Junru Wu, Le Yan, Jiaming Shen, Tianqi Liu, Jialu Liu, Donald Metzler, Xuanhui Wang, and Michael Bendersky. 2024. Large Language Models are Effective Text Rankers with Pairwise Ranking Prompting. https://doi.org/10.48550/arXiv.2306.17563 arXiv:2306.17563 [cs].

[59] Devendra Sachan, Mike Lewis, Mandar Joshi, Armen Aghajanyan, Wen-tau Yih, Joelle Pineau, and Luke Zettlemoyer. 2022. Improving Passage Retrieval with Zero-Shot Question Generation. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Abu Dhabi, United Arab Emirates, 3781–3797. https://doi.org/10.18653/v1/2022.emnlp-main.249

[60] Nihar B. Shah and Martin J. Wainwright. 2016. Simple, Robust and Optimal Ranking from Pairwise Comparisons. http://arxiv.org/abs/1512.08949 arXiv:1512.08949 [cs, math, stat].

[61] Shreya Shankar, Tristan Chambers, Tarak Shah, Aditya G. Parameswaran, and Eugene Wu. 2024. DocETL: Agentic Query Rewriting and Evaluation for Complex Document Processing. https://doi.org/10.48550/arXiv.2410.12189 arXiv:2410.12189 [cs].

[62] Weiwei Sun, Lingyong Yan, Xinyu Ma, Shuaiqiang Wang, Pengjie Ren, Zhumin Chen, Dawei Yin, and Zhaochun Ren. 2023. Is ChatGPT Good at Search? Investigating Large Language Models as Re-Ranking Agents. https://arxiv.org/abs/2304.09542v2

[63] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A Heterogenous Benchmark for Zero-shot Evaluation of Information Retrieval Models. https://doi.org/10.48550/arXiv.2104.08663 arXiv:2104.08663 [cs].

[64] James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. FEVER: a Large-scale Dataset for Fact Extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, Marilyn Walker, Heng Ji, and Amanda Stent (Eds.). Association for Computational Linguistics, New Orleans, Louisiana, 809–819. https://doi.org/10.18653/v1/N18-1074

[65] P. Viola and M. Jones. 2001. Rapid object detection using a boosted cascade of simple features. In *Proceedings of the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition. CVPR 2001*, Vol. 1. IEEE Comput. Soc, Kauai, HI, USA, I–511–I–518. https://doi.org/10.1109/CVPR.2001.990517

[66] Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2024. Text Embeddings by Weakly-Supervised Contrastive Pre-training. http://arxiv.org/abs/2212.03533 arXiv:2212.03533 [cs].

[67] WilliamDAssafMSFT. 2024. Intelligent Applications - Azure SQL Database. https://learn.microsoft.com/en-us/azure/azure-sql/database/ai-artificial-intelligence-intelligent-applications?view=azuresql

[68] Jeff Wu, Long Ouyang, Daniel M. Ziegler, Nisan Stiennon, Ryan Lowe, Jan Leike, and Paul Christiano. 2021. Recursively Summarizing Books with Human Feedback. https://doi.org/10.48550/arXiv.2109.10862 arXiv:2109.10862 [cs].

[69] Shirley Wu, Shiyu Zhao, Michihiro Yasunaga, Kexin Huang, Kaidi Cao, Qian Huang, Vassilis N. Ioannidis, Karthik Subbian, James Zou, and Jure Leskovec. 2024. STaRK: Benchmarking LLM Retrieval on Textual and Relational Knowledge Bases. https://arxiv.org/abs/2404.13207v2

[70] Navid Yaghmazadeh, View Profile, Yuepeng Wang, View Profile, Isil Dillig, View Profile, Thomas Dillig, and View Profile. 2017. SQLizer: query synthesis from natural language. *Proceedings of the ACM on Programming Languages* 1, OOPSLA (Oct. 2017), 1–26. https://doi.org/10.1145/3133887 Publisher: Association for Computing Machinery.

[71] Tao Yu, Zifan Li, Zilin Zhang, Rui Zhang, and Dragomir Radev. 2018. TypeSQL: Knowledge-based Type-Aware Neural Text-to-SQL Generation. https://doi.org/10.48550/arXiv.1804.09769 arXiv:1804.09769 [cs].

[72] Murong Yue, Jie Zhao, Min Zhang, Liang Du, and Ziyu Yao. 2024. Large Language Model Cascades with Mixture of Thoughts Representations for Cost-efficient Reasoning. http://arxiv.org/abs/2310.03094 arXiv:2310.03094 [cs].

[73] John M Zelle and Raymond J Mooney. 1996. 1996-Learning to Parse Database Queries Using Inductive Logic Programming. (1996).

[74] Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. HellaSwag: Can a Machine Really Finish Your Sentence? https://doi.org/10.48550/arXiv.1905.07830 arXiv:1905.07830 [cs].

[75] Bin Zhang, Yuxiao Ye, Guoqing Du, Xiaoru Hu, Zhishuai Li, Sun Yang, Chi Harold Liu, Rui Zhao, Ziyue Li, and Hangyu Mao. 2024. Benchmarking the Text-to-SQL Capability of Large Language Models: A Comprehensive Evaluation. https://doi.org/10.48550/arXiv.2403.02951 arXiv:2403.02951 [cs].

[76] Honglei Zhuang, Zhen Qin, Kai Hui, Junru Wu, Le Yan, Xuanhui Wang, and Michael Bendersky. 2024. Beyond Yes and No: Improving Zero-Shot LLM Rankers via Scoring Fine-Grained Relevance Labels. http://arxiv.org/abs/2310.14122 arXiv:2310.14122 [cs].